

Complete - RL Practical

Now let's build a very simple practical.

Practical: Smart Warehouse Robot using Q-Learning

Story

A robot moves inside a warehouse.

Its objective:

Reach the charging station using the best possible route.

The robot can:

UP

DOWN

LEFT

RIGHT

The agent learns which action is best in each state.

Step 1 — Environment

We'll use this simple grid:

S0	S1	S2	S3	
S4	S5	S6	S7	
S8	S9	S10	S11	
S12	S13	S14	GOAL	

Start:

S0

Goal:

S15

Actions:

0 = LEFT

1 = RIGHT

2 = UP

3 = DOWN

Part 3 — External Dataset

Create an Excel file named:

RL_Warehouse_Robot.xlsx

The Excel file should contain:

State	Action	Next_State	Reward	Done
0	LEFT	0	-1	0
0	RIGHT	1	-1	0
0	UP	0	-1	0
0	DOWN	4	-1	0
1	LEFT	0	-1	0
1	RIGHT	2	-1	0
1	UP	1	-1	0
1	DOWN	5	-1	0
2	LEFT	1	-1	0
2	RIGHT	3	-1	0
2	UP	2	-1	0
2	DOWN	6	-1	0
3	LEFT	2	-1	0
3	RIGHT	3	-1	0
3	UP	3	-1	0
3	DOWN	7	-1	0
4	LEFT	4	-1	0
4	RIGHT	5	-1	0
4	UP	0	-1	0
4	DOWN	8	-1	0
5	LEFT	4	-1	0
5	RIGHT	6	-1	0
5	UP	1	-1	0
5	DOWN	9	-1	0
6	LEFT	5	-1	0
6	RIGHT	7	-1	0
6	UP	2	-1	0

6	DOWN	10	-1	0
7	LEFT	6	-1	0
7	RIGHT	7	-1	0
7	UP	3	-1	0
7	DOWN	11	-1	0
8	LEFT	8	-1	0
8	RIGHT	9	-1	0
8	UP	4	-1	0
8	DOWN	12	-1	0
9	LEFT	8	-1	0
9	RIGHT	10	-1	0
9	UP	5	-1	0
9	DOWN	13	-1	0
10	LEFT	9	-1	0
10	RIGHT	11	-1	0
10	UP	6	-1	0
10	DOWN	14	-1	0
11	LEFT	10	-1	0
11	RIGHT	11	-1	0
11	UP	7	-1	0
11	DOWN	15	100	1
12	LEFT	12	-1	0
12	RIGHT	13	-1	0
12	UP	8	-1	0
12	DOWN	12	-1	0
13	LEFT	12	-1	0
13	RIGHT	14	-1	0
13	UP	9	-1	0
13	DOWN	13	-1	0
14	LEFT	13	-1	0
14	RIGHT	15	100	1
14	UP	10	-1	0
14	DOWN	14	-1	0
15	LEFT	15	0	1
15	RIGHT	15	0	1
15	UP	15	0	1
15	DOWN	15	0	1

Upload above file from external dataset as per given

Explain the dataset

This is very important.

The dataset isn't like your previous supervised-learning datasets.

There isn't:

Input → Correct Answer

Instead, it describes:

Current State



Action



Next State



Reward

For example:

State = 14

Action = RIGHT

Next State = 15

Reward = 100

Done = 1

Meaning:

Robot is at State 14 → moves RIGHT → reaches the goal → receives +100 reward → episode ends.